

FAGARAS, Romania

WMO: 152350

Lat: 45.8364N

Lon: 24.9353E

Elev: 1408

StdP: 13.96

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.2	6.4	-2.3	5.1	0.4	3.9	7.1	7.4	22.6	31.0	18.8	35.0	0.7	70	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.4	87.0	68.9	84.2	67.5	81.5	66.6	72.7	81.3	70.6	79.9	68.9	78.0	4.1	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.3	117.3	75.8	67.8	107.8	73.4	66.1	101.2	72.2	37.3	80.8	35.4	80.2	33.9	77.6	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.2	16.3	13.1	-7.9	91.5	8.9	3.3	-14.3	93.8	-19.5	95.8	-24.5	97.6	-30.9	100.0		
(5)			-10.0	76.5	8.8	1.9	-16.4	77.9	-21.5	79.0	-26.4	80.0	-32.8	81.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.6	26.8	32.3	39.6	49.7	58.2	65.2	67.2	67.1	59.0	49.1	39.1	28.9		
	DBStd	16.05	9.20	8.69	7.24	7.60	5.70	5.09	4.46	4.86	6.89	6.53	7.88	9.75		
	HDD50	2740	720	496	325	97	8	0	0	0	8	97	335	654		
	HDD65	6276	1185	916	787	460	217	57	30	35	200	493	777	1119		
	CDD50	2228	0	0	4	86	263	457	533	530	278	69	8	0		
	CDD65	288	0	0	0	0	6	64	98	100	20	0	0	0		
	CDH74	3493	0	0	0	44	178	709	1004	1179	363	16	0	0		
	CDH80	917	0	0	0	2	21	161	267	377	89	0	0	0		
(14) Wind	WSAvg	4.0	3.3	4.1	5.1	5.4	4.3	3.8	3.6	3.3	4.0	3.7	3.4	3.7		
(15) Precipitation	PrecAvg	27.8	1.3	1.2	1.5	2.2	3.4	4.1	4.0	2.9	2.4	2.0	1.5	1.5		
	PrecMax	38.3	2.8	2.3	2.7	4.4	7.1	8.1	7.4	6.2	5.8	3.9	3.6	3.9		
	PrecMin	17.4	0.3	0.2	0.3	0.7	1.3	1.1	1.0	0.6	0.4	0.1	0.0	0.2		
	PrecStd	5.0	0.6	0.6	0.6	0.9	1.6	1.8	1.6	1.7	1.2	1.0	0.8	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.0	59.8	67.9	78.3	82.7	88.0	89.6	91.2	86.9	75.6	66.7	50.1		
		MCWB	43.5	49.9	52.2	58.9	62.4	70.6	70.0	70.8	67.5	59.7	54.5	44.4		
	2%	DB	44.0	53.0	61.9	74.2	78.4	84.3	86.2	87.3	82.3	72.0	60.4	45.8		
		MCWB	41.0	45.6	49.3	56.9	62.0	68.2	69.5	68.4	64.6	57.7	52.4	42.8		
	5%	DB	39.8	48.2	56.8	68.7	75.3	81.1	82.9	84.1	78.1	67.4	55.8	42.2		
		MCWB	37.5	43.4	46.7	55.4	61.1	66.8	68.9	67.0	63.1	56.1	50.2	40.3		
	10%	DB	36.8	44.2	52.0	64.4	71.9	78.2	80.0	80.9	73.7	62.6	51.5	39.3		
		MCWB	35.5	40.9	44.9	53.4	59.9	65.9	68.0	66.3	61.8	54.5	47.2	38.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.9	50.7	53.7	61.5	67.5	74.5	76.8	75.0	70.4	63.3	56.4	45.5		
		MCDB	47.5	58.3	67.2	71.4	77.2	83.5	79.3	83.0	80.1	70.6	63.1	48.3		
	2%	WB	41.2	46.8	50.5	58.9	64.8	70.9	72.9	72.0	67.4	59.9	53.4	43.2		
		MCDB	43.9	52.1	59.9	70.7	73.8	79.3	80.3	82.7	78.1	68.9	58.9	46.0		
	5%	WB	38.0	44.0	48.3	56.7	62.9	68.8	70.9	69.5	65.2	57.7	50.7	40.6		
		MCDB	39.4	47.6	55.7	66.9	71.5	77.1	79.7	79.7	74.0	65.6	55.0	42.4		
	10%	WB	35.7	41.4	45.8	54.7	61.1	67.2	69.2	67.7	63.2	55.5	47.6	38.5		
		MCDB	36.9	44.1	51.0	62.8	69.3	75.5	78.0	77.5	71.6	61.3	50.9	39.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.5	17.2	18.6	21.9	21.3	21.5	22.4	24.4	22.9	22.8	17.3	12.9		
		MCDBR	15.9	23.0	29.6	31.7	29.3	27.1	28.8	30.8	31.6	33.7	25.9	15.7		
	5% WB	MCWBR	12.4	15.6	17.2	15.5	14.1	13.8	14.5	14.5	16.1	18.5	18.5	11.5		
		MCWBR	15.1	21.9	26.3	27.4	25.3	24.9	27.0	28.5	28.9	27.4	24.1	13.7		
(40) Clear-Sky Solar Irradiance	taub		0.305	0.333	0.375	0.432	0.415	0.413	0.431	0.441	0.397	0.359	0.334	0.306		
		taud	2.332	2.275	2.234	2.118	2.234	2.275	2.232	2.200	2.294	2.391	2.406	2.371		
	Ebn at Noon		250	264	268	261	271	271	264	256	258	252	238	237		
		Edh at Noon	24	31	38	47	43	41	43	43	36	28	23	21		
(44) All-Sky Solar Radiation	RadAvg		433	697	1039	1363	1668	1806	1820	1657	1203	831	482	345		
	RadStd		48	84	113	146	161	161	156	143	123	104	53	41		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (78 neighbors)	+1.08	N/A	N/A	+1.30	N/A	N/A	-230	-313	+137	+43

Nomenclature: See separate page